CANopen Dynamic PDO Mapping Configuration Guide

1. Overview & Setup

This document provides a technical guide for modifying **PDO mappings** dynamically via **SDO (Service Data Object)** messages on CANopen slave devices.

1.1 Node & Addressing Assumptions

- **Slave Node ID (Node-ID)**: `0x01` (Decimal 1)
- **SDO Request COB-ID**: `0x600 + Node-ID = 0x601`
- **SDO Response COB-ID**: `0x580 + Node-ID = 0x581`
- **Mapping Parameters Index**: `0x1A00` to `0x1A03` (Corresponding to TPDO1 ~ TPDO4)
- **Communication Parameters Index**: `0x1800` to `0x1803` (Corresponding to TPDO1 ~ TPDO4)

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2. SDO Parameter Mapping Conversion Rule

According to the CANopen specification (CiA 301), a PDO mapping entry is a **32-bit unsigned integer (UINT32)** structured as follows:

$$extMappingEntry(32-bit) = extIndex(16bits) \parallel extSubindex(8bits) \parallel extBitLength(8bits)$$

2.1 Bit Decomposition

- **High 16 Bits (Bits 31~16)**: Target Object Dictionary **Index**
- **Middle 8 Bits (Bits 15~8)**: Target Object Dictionary **Subindex**
- **Low 8 Bits (Bits 7~0)**: Data type **Bit Length**
  - `float` (Single-precision floating point, 32 bits) $ightarrr$ `0x20` (32 bits)
  - `uint16` / `int16` (16-bit integer) $ightarrr$ `0x10` (16 bits)
  - `uint8` / `int8` (8-bit integer) $ightarrr$ `0x08` (8 bits)

2.2 Little-Endian Byte Order Rule

CANopen transfers multi-byte values in **Little-Endian** format (least significant byte first).

Conversion Example:

To map `Index = 0x300B`, `Subindex = 0x01`, Data Type = `float` (32 bit / `0x20`):

1. **Combine into 32-bit HEX**: `0x300B0120`
2. **Decompose into bytes (MSB to LSB)**: `0x30`, `0x0B`, `0x01`, `0x20`
3. **Reorder to Little-Endian CAN payload** (Bytes D4, D5, D6, D7):
   - `D4` (LSB) = `0x20` (Bit length)
   - `D5` = `0x01` (Subindex)
   - `D6` = `0x0B` (Index LSB)
   - `D7` (MSB) = `0x30` (Index MSB)

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3. Standard 6-Step PDO Reconfiguration Sequence

To reconfigure PDO mappings dynamically, follow this standard sequence:

1. **Disable PDO**: Set Bit 31 of COB-ID at `0x1800+n Sub01` to `1`.
2. **Clear Mapping Count**: Write `0` to `0x1A00+n Sub00`.
3. **Write Mapping Entry 1**: Write mapping entry to `0x1A00+n Sub01`.
4. **Write Mapping Entry 2**: Write mapping entry to `0x1A00+n Sub02`.
5. **Set Mapping Count**: Write the total number of mapped items (e.g. `2`) to `0x1A00+n Sub00`.
6. **Enable PDO**: Clear Bit 31 of COB-ID at `0x1800+n Sub01` back to `0`.

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4. Complete Configuration Messages

4.1 TPDO1 Configuration (Channel 1 Voltage & Current ➡️ PDO1)

- **Target Objects**: `0x300B Sub 0x01` (Ch1 Voltage float) + `0x300B Sub 0x02` (Ch1 Current float)
- **TPDO1 Comm Index**: `0x1800` | **TPDO1 Mapping Index**: `0x1A00`

Step	Operation	COB-ID	DLC	Data (HEX)	Description
1	Disable TPDO1	`0x601`	8	`23 00 18 01 81 01 00 80`	Write 0x1800 Sub01 = 0x80000181 (Bit31=1)
-	*Slave Response*	`0x581`	8	`60 00 18 01 00 00 00 00`	Success
2	Clear Mapping Count	`0x601`	8	`2F 00 1A 00 00 00 00 00`	Write 0x1A00 Sub00 = 0x00

Step	Operation	COB-ID	DLC	Data (HEX)	Description
-	*Slave Response*	`0x581`	8	`60 00 1A 00 00 00 00 00`	Success
3	Map Entry 1 (Ch1 Voltage)	`0x601`	8	`23 00 1A 01 20 01 0B 30`	Write 0x1A00 Sub01 = 0x300B0120
-	*Slave Response*	`0x581`	8	`60 00 1A 01 00 00 00 00`	Success
4	Map Entry 2 (Ch1 Current)	`0x601`	8	`23 00 1A 02 20 02 0B 30`	Write 0x1A00 Sub02 = 0x300B0220
-	*Slave Response*	`0x581`	8	`60 00 1A 02 00 00 00 00`	Success
5	Set Mapping Count = 2	`0x601`	8	`2F 00 1A 00 02 00 00 00`	Write 0x1A00 Sub00 = 0x02
-	*Slave Response*	`0x581`	8	`60 00 1A 00 00 00 00 00`	Success
6	Enable TPDO1	`0x601`	8	`23 00 18 01 81 01 00 00`	Write 0x1800 Sub01 = 0x00000181 (Bit31=0)
-	*Slave Response*	`0x581`	8	`60 00 18 01 00 00 00 00`	Success

4.2 TPDO2 Configuration (Channel 2 Voltage & Current ➡️ PDO2)

- **Target Objects**: `0x310B Sub 0x01` (Ch2 Voltage float) + `0x310B Sub 0x02` (Ch2 Current float)
- **TPDO2 Comm Index**: `0x1801` | **TPDO2 Mapping Index**: `0x1A01`

Step	Operation	COB-ID	DLC	Data (HEX)	Description
1	Disable TPDO2	`0x601`	8	`23 01 18 01 82 01 00 80`	Write 0x1801 Sub01 = 0x80000281
2	Clear Mapping Count	`0x601`	8	`2F 01 1A 00 00 00 00 00`	Write 0x1A01 Sub00 = 0x00
3	Map Entry 1 (Ch2 Voltage)	`0x601`	8	`23 01 1A 01 20 01 0B 31`	Write 0x1A01 Sub01 = 0x310B0120
4	Map Entry 2 (Ch2 Current)	`0x601`	8	`23 01 1A 02 20 02 0B 31`	Write 0x1A01 Sub02 = 0x310B0220

Step	Operation	COB-ID	DLC	Data (HEX)	Description
5	Set Mapping Count = 2	`0x601`	8	`2F 01 1A 00 02 00 00 00`	Write 0x1A01 Sub00 = 0x02
6	Enable TPDO2	`0x601`	8	`23 01 18 01 82 01 00 00`	Write 0x1801 Sub01 = 0x00000281

4.3 TPDO3 Configuration (Channel 3 Voltage & Current ➡️ PDO3)

- **Target Objects**: `0x320B Sub 0x01` (Ch3 Voltage float) + `0x320B Sub 0x02` (Ch3 Current float)
- **TPDO3 Comm Index**: `0x1802` | **TPDO3 Mapping Index**: `0x1A02`

Step	Operation	COB-ID	DLC	Data (HEX)	Description
1	Disable TPDO3	`0x601`	8	`23 02 18 01 83 01 00 80`	Write 0x1802 Sub01 = 0x80000381
2	Clear Mapping Count	`0x601`	8	`2F 02 1A 00 00 00 00 00`	Write 0x1A02 Sub00 = 0x00
3	Map Entry 1 (Ch3 Voltage)	`0x601`	8	`23 02 1A 01 20 01 0B 32`	Write 0x1A02 Sub01 = 0x320B0120
4	Map Entry 2 (Ch3 Current)	`0x601`	8	`23 02 1A 02 20 02 0B 32`	Write 0x1A02 Sub02 = 0x320B0220
5	Set Mapping Count = 2	`0x601`	8	`2F 02 1A 00 02 00 00 00`	Write 0x1A02 Sub00 = 0x02
6	Enable TPDO3	`0x601`	8	`23 02 18 01 83 01 00 00`	Write 0x1802 Sub01 = 0x00000381

4.4 TPDO4 Configuration (Channel 5 Voltage & Current ➡️ PDO4)

- **Target Objects**: `0x340B Sub 0x01` (Ch5 Voltage float) + `0x340B Sub 0x02` (Ch5 Current float)
- **TPDO4 Comm Index**: `0x1803` | **TPDO4 Mapping Index**: `0x1A03`

Step	Operation	COB-ID	DLC	Data (HEX)	Description
1	Disable TPDO4	`0x601`	8	`23 03 18 01 84 01 00 80`	Write 0x1803 Sub01 = 0x80000481
2	Clear Mapping Count	`0x601`	8	`2F 03 1A 00 00 00 00 00`	Write 0x1A03 Sub00 = 0x00
3	Map Entry 1 (Ch5 Voltage)	`0x601`	8	`23 03 1A 01 20 01 0B 34`	Write 0x1A03 Sub01 = 0x340B0120
4	Map Entry 2 (Ch5 Current)	`0x601`	8	`23 03 1A 02 20 02 0B 34`	Write 0x1A03 Sub02 = 0x340B0220
5	Set Mapping Count = 2	`0x601`	8	`2F 03 1A 00 02 00 00 00`	Write 0x1A03 Sub00 = 0x02
6	Enable TPDO4	`0x601`	8	`23 03 18 01 84 01 00 00`	Write 0x1803 Sub01 = 0x00000481

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5. Quick Reference: SDO Command Bytes

Command Byte (HEX)	Length	Type	Purpose
`0x2F`	1 Byte	Expedited	Writing Sub00 (Mapping count, e.g. `0x02`)
`0x2B`	2 Bytes	Expedited	Writing 16-bit parameters
`0x23`	4 Bytes	Expedited	Writing 32-bit mapping entries & COB-IDs